

industry in computing performance, gaming, imaging, multimedia, and connectivity innovations. It integrates the cutting-edge Armv9 CPU archi-

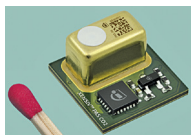


ture for a true flagship experience. The octa-core CPU has one ultra-Cortex-X2 core operating at up to 3.05GHz, three performance A710 cores operating at up to 2.85GHz, and four efficiency Cortex-A510 cores. With the integrated LPDDR5X supporting up to 7500Mbps, along with an 8MB L3 cache and 6MB system cache, the chip is built for the massive bandwidth demands of the mobile market.

MediaTek
www.mediatek.com

CO₂ sensor

To meet the rising demand for sensors that measure the CO₂ concentration

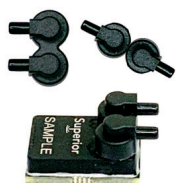


in the air, Infineon Technologies AG is launching an innovative CO₂ sensor known as the XENSIV PAS CO2 that can be used to monitor indoor air quality and reduce energy costs. Suitable applications include ventilation, air-conditioning systems, portable room air monitoring devices, and smart speakers. Based on the innovative PAS (photoacoustic spectroscopy) technology, the sensor has an integrated microcontroller that directly converts the CO₂ measurement into a ppm value, which is available via three interfaces: the serial I²C, UART, and PWM.

Infineon Technologies
www.infineon.com

Pressure sensors

The new ND series absolute mid-pressure sensors and the differential and gauge mid-pressure sensors enable manufacturers to cost-effectively develop more accurate and stable products

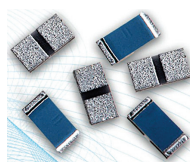


for numerous industrial equipment applications operating at low- to mid-pressure ranges. The absolute sensors support pressures up to 150psia with market leading accuracy and long-term stability of 0.10%, while the differential/gauge sensors support multi-range pressures from 0.5psi to 150psi with market leading accuracy and long-term stability of 0.05%. These are said to offer manufacturers a 2x advantage over other solutions.

Superior Sensor Technology
www.superiorsensors.com

HPDC resistors

The new high power density chip (HPDC) resistors are optimised for power management, actuator drive, and heating applications which benefit

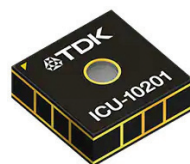


from the enhanced heat transfer from element to terminal. The use of such high power density components saves PCB area and boosts reliability by restricting the temperature rise in the component hotspot. The resistors use an aluminium nitride ceramic substrate and feature large area terminations for improved thermal contact with a PCB. As a result, heat generated in the resistor element is effectively conducted away, reducing the temperature rise and enabling the HPDC resistors to offer a high power density solution with ratings of 2.4W in 1206 and 3.5W in 2512 footprints.

TTElectronics
www.ttelectronics.com

Ultrasonic ToF sensors

The Chirp ICU-10201 and ICU-20201, two new high-performance, ultra-low-



power integrated ultrasonic Time-of-Flight (ToF) sensors for short- and long-range detection include a powerful on-chip processor with higher computational power. The enhanced processing capabilities allow a wide

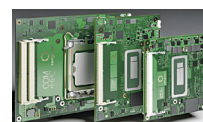
range of application algorithms to fit and to run on-chip, completely off-loading the system MCU. Compared to the previous generation, the ICU-10201 and ICU-20201 have 10 times faster DSP with HW multiplier for higher calculation power, larger memory, and a fast host interface (SPI). The sensors also support a wide range of IO supply voltages with range measurements to targets up to 5m (for ICU-20201) and 1.2m (for ICU-10201) away.

TDK Corporation
www.tdk.com

BOARDS & MODULES

Computer-on-modules

The 10 new COM-HPC and COM Express computer-on-modules feature



the latest high-performance cores from Intel. The new modules in COM-HPC Size A and C as well as COM Express Type 6 form factors offer major performance gains and improvements for the world of embedded and edge computing systems. Offering up to 14 cores/20 threads on BGA and 16 cores/24 threads on desktop variants (LGA mounted), 12th Gen Intel Core processors provide a quantum leap in multitasking and scalability levels. Next-gen IoT and edge applications benefit from up to 6 or 8 (BGA/LGA) optimised Performance-cores (P-cores) plus up to 8 low-power Efficient-cores (E-cores) and DDR5 memory support to accelerate multithreaded applications and execute background tasks more efficiently.

congatec
www.congatec.com

IoT module

The AWS IoT ExpressLink module, ESP32-C3-MINI-1-N4-A, is based on the ESP32-C3 SoC and offers Wi-Fi + Bluetooth LE 5.0 connectivity, thus providing any host MCUs with out-of-the-box, seamless, AWS cloud connectivity. The module offers a simple